

Aim

To design and deploy a trust-gated, hybrid-RAG-powered agentic framework on real Azure Kubernetes/AWS infrastructure that makes carbon-aware microservice scheduling decisions reliably in real time — combining LLM adaptability with a deterministic MILP safety fallback — validated against live workload traffic and real Electricity Maps carbon-intensity data across two Azure/AWS regions.

Abstract

- Cloud data centers are a major and growing source of carbon emissions, and existing carbon-aware scheduling relies on rigid mathematical solvers that struggle with dynamic microservice dependencies.
- Autonomous LLM agents offer adaptive scheduling decisions but risk hallucination and unsafe actions if deployed without safeguards.
- This project proposes a Hybrid RAG (graph + vector) agentic system where a multi-agent LLM pipeline proposes carbon-aware scheduling actions, gated by a real-time trust score (RAGAS + rule-based guardrails).
- Low-trust decisions automatically fall back to a deterministic Mixed Integer Linear Programming(MILP) solver, ensuring Service Level Objective(SLO) compliance is never sacrificed for adaptability.
- The system is deployed on real AKS/AWS running an actual DeathStarBench microservice application under live load, with real-time carbon intensity pulled directly from the Electricity Maps API for two Azure regions — comparing unconstrained-agent, MILP-only, and trust-gated hybrid configurations against genuine infrastructure behavior rather than a simulated environment.
- Results quantify carbon savings, SLO violation rates, and fallback trigger frequency, demonstrating whether trust-gating preserves most of the LLM's adaptive benefit while eliminating its unsafe-decision risk.

Literature Review

Paper	Objective	Methodology	Pros	Findings
Dependency-Aware Online Microservice Re-Scheduling for Adaptive Resources Co-Optimization in Edge Networks – IEEE Transactions on Services Computing (Nov.-Dec. 2025)	Co-optimize edge resources via microservice re-scheduling	D3QN reinforcement learning & dependency graphs	Pros: Balances resources; cuts energy use. Cons: High training/computational overhead.	Improved latency by 1.85% and energy efficiency by 6.45%.
Towards sustainable computing: Influences, trade-offs, and techniques for energy-aware resource management in the cloud-edge continuum – Elsevier - Renewable and Sustainable Energy Reviews , November 2026	Optimize energy-aware management in cloud-edge systems.	Systematic review & mathematical modeling.	Pros: Comprehensive framework. Cons: Lacks live, large-scale empirical data.	Identified key trade-offs; models show up to 60–99% energy savings.

Paper	Objective	Methodology	Pros	Findings
CARA: Adaptive scheduling for carbon-efficient workload orchestration in distributed cloud systems – Elsevier - Sustainable Computing: Informatics and Systems , September 2026	Schedule workloads using spatio-temporal shifting	Greedy sub-optimal scheduling in Kubernetes based on CO2 data.	Pros: Unifies spatial & temporal shifting. Cons: Greedy algorithms may miss the global optimum	Successfully filled the gap for combined spatio-temporal optimization.
A two-layer MILP matheuristic approach for multi-mode resource-constrained multi-project scheduling problem considering uncertain project release time – Elsevier - Expert Systems with Applications , September 2026	Schedule multi-projects with uncertain release times.	Two-layer MILP matheuristic approach	Pros: Handles complex uncertainties well. Cons: MILP scales poorly on massive datasets.	Successfully solved multi-mode resource-constrained scheduling issues.
Energy-aware elastic scaling algorithm for microservices in Kubernetes clouds – Elsevier - Journal of Network and Computer Applications , October 2025	Adjust Kubernetes HPA thresholds to save energy.	Moving average smoothing & data analysis.	Pros: Cuts peak-load energy by 50%. Cons: Moving averages lag during sudden spikes.	Reduced instantiated nodes by 15% while improving average latency by 15%.

Research Gap

- **Topological blindness:** Existing agentic AIOps systems use flat/unstructured context and lack real-time awareness of microservice dependency structure, causing carbon-driven decisions to trigger latency or network cost regressions.
- **Absence of bounded trust guarantees:** Current LLM-driven infrastructure agents lack real-time, quantified confidence checks before executing actions, risking cascading failures from hallucinated decisions.
- **Static vs. dynamic optimization:** Traditional carbon-aware schedulers treat workloads as binary (delayable/static) and don't adapt scheduling dynamically to fine-grained, real-time carbon/latency trade-offs.
- **Fragmented observability:** System infrastructure metrics and LLM behavioral/reasoning metrics are typically monitored in separate tools, with no unified trace correlating carbon decisions to agent confidence.

Objectives

- Build a structural dependency-graph representation (Neo4j) of a real microservice application to give LLM agents topological context.
- Implement a Hybrid RAG pipeline (Neo4j + Qdrant) that fuses graph-structured and unstructured operational context for agent reasoning.
- Design a multi-agent LLM pipeline (LangGraph) that proposes carbon- and SLO-aware scheduling actions.
- Develop a quantitative trust-scoring mechanism (RAGAS + NeMo Guardrails) that gates agent actions before execution.
- Implement a deterministic MILP fallback (PuLP) triggered whenever agent trust falls below a safety threshold.
- Instrument the full decision pipeline with OpenTelemetry tracing for reproducible observability and analysis.
- Empirically compare unconstrained-agent, MILP-only, and trust-gated hybrid configurations on carbon reduction and Service Level Objective(SLO) compliance using real trace data.



AZURE — Resource Group

 **AKS Cluster (free control plane, Free tier)**
2-3 worker node VMs (e.g. B2s)

DeathStarBench
(real pods, real traffic via wrk/hey)



Neo4j / Qdrant

-  - Neo4j (dependency graph)
-  - Qdrant (log/context retrieval)

Controller pod

- LangGraph agent flow (calls Ollama)
- RAGAS + NeMo Guardrails scoring
- PuLP MILP fallback
- K8s API calls to actually reschedule pods



OTel Collector + Jaeger (traces every stage)



Data Center: **Electricity Maps API**  Region 



(Data center → Provider: azure → Region query for whichever 2 Azure regions you deploy into, e.g. West US 2 + East US)

Electricity Maps: carbon intensity for your 2 Azure regions

Hybrid RAG: Neo4j (dependency context) + Qdrant (log context)

LangGraph agents: Profiler → Predictor → Action (via Ollama on laptop)

Guardrail: RAGAS + NeMo Guardrails

Trust Score?

trusted

not trusted

K8s API: reschedule real pods

PuLP MILP solves placement, then same K8s API

Measure real latency (wrk/hey) + real carbon (Electricity Maps)

Log to CSV/JSON + OTel trace in Jaeger

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